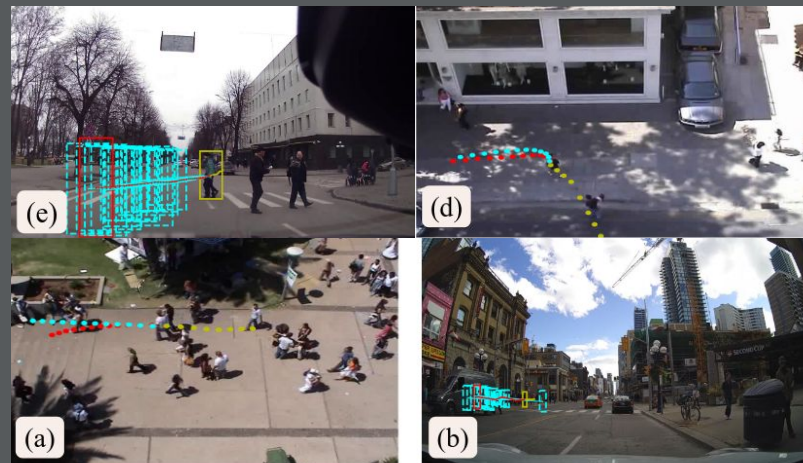


Bridging Visual Semantics and Motion Dynamics: CLIP-Augmented Pedestrian Trajectory Prediction

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Problem Description

- 1) VLMs have ‘weak’ understanding of world rules
 - a) Temporal dynamics
 - b) Potential hazards
 - c) Intent
 - d) World model design*
- 2) VLAs lack of grounding behind causal relationships of ‘seeing’ and ‘doing’
- 3) Traditional approaches (model-based, probabilistic, graph-based) implicitly understand world rules, but don’t have explicit semantics...



Note: ground truth is red, cyan is prediction, and yellow is observed (past up to prediction step)

Methodology and Experimental Setup

1) Component descriptions:

- a) SGNet: Stepwise Goal-Driven Networks for Trajectory Prediction
 - i) Implicit temporal motion modeling w/ GRU
 - ii) Goal-conditioned prediction (stepwise goal estimator)
 - iii) Stochastic future modeling w/ CVAE (dynamic uncertainty)
- b) CLIP:
 - i) Frozen ViT-B/32 image encoder producing 512-dim embeddings from per-timestep pedestrian image crops

2) Integration strategies:

- a) Naive control: pure concatenation CLIP embeddings to bounding box inputs
- b) Cross attention: GRU hidden state queries corresponding CLIP visual token, fusing visual semantic features into motion representations

3) Setup:

- a) Dataset - PIE [\[2\]](#) Set01 (4 captures, 1920x1080) | Train/Validation/Test Split - Sliding Windows*
- b) Input - 15 observed frames | Output - 45 future frames | Epochs - 20 | Batch Size - 16 | Learning Rate - $1e-3$ | Optimizer - Adam | GRU Hidden Size - 512 | CVAE Latent Dim - 32 | $K = 20$ hypotheses

Results and Analysis

- Baseline still beats architectural changes to integrate CLIP embeddings...?
 - Conservative training setup
 - Hyperparameter tuning
 - Limited dataset + (relatively) noisy signal injection
- Note: error metrics are in pixels

⇒ The goal was not to beat the baseline...

⇒ Wanted to recover visual embeddings (and bonus is to match or beat performance)!

Model	MSE_05	MSE_10	MSE_15	Best Epoch
Baseline	22.09	50.86	124.24	20
Concat	32.54	82.67	201.80	20
Cross-attn	32.52	79.74	183.53	19

Model	MSE_15	ADE (px)	FDE (px)	CMSE	CFMSE
Baseline	126.58	7.20	11.77	93.80	302.19
Cross-attn	176.97	8.55	14.41	140.98	423.39
Concat	196.63	9.10	15.41	159.91	480.49

Discussion

1) Implications and Research Context

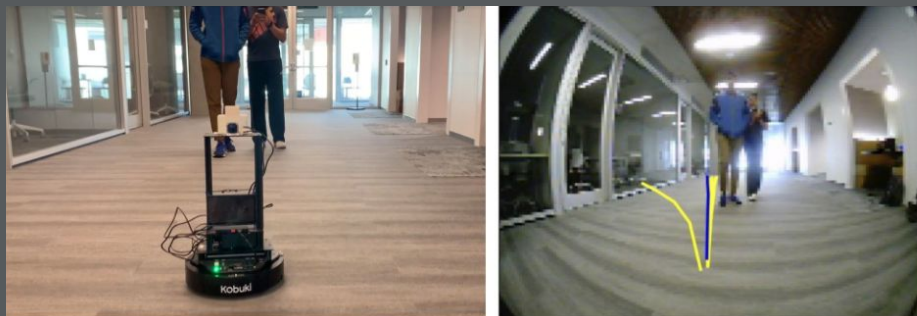
- a) Latent space representations for (visual) reasoning behind trajectory prediction
- b) Potential for deriving grounded uncertainty (what visual cues link to model failure?)
- c) Introduces a (separate) ‘dynamics prediction simulator’ into a VLM-like pipeline
- d) Departure from “dreaming”?

2) Limitations

- a) CLIP embedding image crops (important!)
- b) Training (dataset, tuning, etc.)
- c) Model integration architecture
- d) Loss of intent supervision (from PIE)

3) Future Work

- a) Full-frame scene tokens
- b) VQA-Style inspections
- c) Testing w/ other VLMs



References

[1] Chuhua Wang, Yuchen Wang, Mingze Xu, and David J. Crandall. "Stepwise Goal-Driven Networks for Trajectory Prediction." arXiv:2103.14107, 2022.

[2] Amir Rasouli, Iuliia Kotseruba, Toni Kunic, and John K. Tsotsos. "PIE: A Large-Scale Dataset and Models for Pedestrian Intention Estimation and Trajectory Prediction." ICCV, 2019.

[3] Dhruv Shah, Ajay Sridhar, Nitish Dashora, Kyle Stachowicz, Kevin Black, Noriaki Hirose, and Sergey Levine. "ViNT: A Foundation Model for Visual Navigation." arXiv:2306.14846, 2023.

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